

PROMPT ENGINEERING

1.What is prompting?

-> Prompting is the process of giving a question, instruction, or command to an AI model so that it can understand the task and generate an appropriate response.

Example

Prompt:

"Explain Neural Networks in simple words."

↓

AI understands the request

↓

AI generates an easy explanation

↓

User receives the response

2.How LLM's Understand Prompt

-> When a user gives a prompt, the LLM first understands the natural language and converts it into tokens. The Transformer then analyzes the relationships and context between those tokens, while the neural network uses its learned parameters to predict the most appropriate next tokens.

->By repeatedly predicting tokens, the model generates a response with proper grammar and context.

Easy Memory Flow

User Prompt

↓

Tokenizer

↓

LLM understands language

↓

Transformer understands context

↓

Neural network uses parameters

↓

Next-token prediction -> Response with proper grammar and context

Super Simple Example

Prompt:

Translate "Hello" into French.

LLM understands:

- Task → Translation
- Input → "Hello"
- Target language → French

Output:

Bonjour

3.Role Prompting

->Role Prompting is the technique of assigning a role (such as teacher, doctor, programmer, or interviewer) to the AI so that it generates responses in that style or area of expertise.

How Does Role Prompting Help?

Without a role:

Question

↓

General Answer

With a role:

Assign Role

↓

Question

↓

AI Adapts Its Style and Perspective

↓

More Relevant Answer

Example

Prompt: Act as an AI professor and explain Neural Networks with a simple example.

->The AI takes the role of an AI professor and provides a teaching-style explanation.

4.Context Prompting

-> Context Prompting is a technique where additional background information or details are provided to the AI so that it can generate a more accurate and relevant response.

In simple words:

-> **You first tell the AI the situation (context)**, then ask your question.

Difference Between Role Prompting and Context Prompting

Role Prompting

Tells AI **who it should be**.

Example: "Act as an AI professor."

Context Prompting

Tells AI **what the situation is**.

Example: "I am a beginner preparing for an exam."

Example

Prompt:

I am a beginner learning AI for the first time.

Explain Neural Networks in simple words with a real-life example.

Here:

- **Context:** "I am a beginner learning AI for the first time."
- **Task:** "Explain Neural Networks in simple words with a real-life example."

-> The AI uses the context to generate an answer that matches the user's level.

5. Structured Prompting

-> Structured Prompting means arranging the instructions, context, and requirements in a proper format so that the AI clearly understands what kind of response is expected.

Example

Structured Prompt:

Role: Act as a college teacher.

Context: I am a beginner learning AI.

Task: Explain Transformers.

Requirements:

- Use simple language.
- Give one real-life example.
- Keep it within 100 words.

->The AI understands each part separately and generates a focused answer.

Easy Memory Trick

->Role Prompting = Who should AI be?

->Context Prompting = What is the situation?

->Structured Prompting = How should the instructions be organized?

ADVANCED PROMPTING

6.Chain-of-Thought Prompting

-> Chain-of-Thought Prompting means asking the AI to think and solve a problem step by step before generating the final answer, which helps improve reasoning and accuracy.

Normal Prompt

↓

Give Answer

Chain-of-Thought Prompt

↓

Think Step by Step-> Give Answer

Example

Prompt:

->Solve this math problem step by step:

If a train travels 60 km in 1 hour, how far will it travel in 3 hours?

AI Response:

Speed = 60 km/hour

Time = 3 hours

Distance = Speed × Time

= 60 × 3

= 180 km

->Final Answer: 180 km.

7.Few-Shot Prompting

-> Few-Shot Prompting is the technique of giving the AI a few examples before the actual task so it can learn the pattern and generate a better response.

Real-Life Analogy

Imagine a teacher says:

"Complete the pattern."

She first gives:

$$2 + 2 = 4$$

$$3 + 3 = 6$$

$$4 + 4 = ?$$

->You understand the pattern from the examples and answer: 8

->Few-Shot Prompting works the same way. The AI learns the expected pattern from the given examples.

How Does Few Shot Prompting Work?

Give 2–3 Examples

↓

AI Learns the Pattern

↓

Ask New Question

↓

AI Follows the Same Pattern

Difference Between Zero-Shot, One-Shot, and Few-Shot

	Type	Meaning
Zero-Shot		No examples are given.
One-Shot		One example is given.
Few-Shot		Two or more examples are given.

Example With Few-Shot

Prompt:

Example 1:

Sentence: "I love this phone."

Sentiment: Positive

Example 2:

Sentence: "The service was terrible."

Sentiment: Negative

Now classify:

Sentence: "The food was delicious."

Sentiment:

AI Response: Positive

->The examples help AI understand exactly what you expect.

8.Multi Model Prompting

->Multimodal Prompting is the technique of providing multiple types of input (text, images, audio, or video) in a prompt to help the AI generate a more accurate and context-aware response.

Connection with Multimodal AI

You already learned:

Multimodal AI can understand multiple types of data.

Now:

Multimodal Prompting is the way we give those multiple types of data to the AI.

Think of it like this:

Multimodal AI

↓

Has the ability to understand text, images, audio, etc.

Multimodal Prompting

↓

The user actually provides text + image + audio, etc., in the prompt.

Example

-> Suppose you upload a picture of a circuit diagram and ask:

Explain how this circuit works in simple words.

The AI:

1. Understands the **image** (circuit diagram).
2. Understands the **text instruction** ("explain in simple words").
3. Combines both and generates the answer.

Here, the prompt contains:

-  **Image**
-  **Text**

-> So, it is a **Multimodal Prompt**.

Applied Generative AI & Intelligent Systems

9. Llama

-> Llama is an open-source Large Language Model developed by Meta. Like ChatGPT, it understands and generates natural language, but because it is open source, developers and companies can download it, fine-tune it, and build their own custom AI systems.

Example

General Llama Model

↓

Fine-Tuned with Medical Data

↓

Medical AI Assistant

Or:

General Llama Model

↓

Connected to Company Documents using RAG



Private Company Chatbot

10.YOLO

-> YOLO is an AI model that can look at an image or video and quickly identify and locate objects like people, cars, dogs, traffic lights, etc.

Human Analogy

Imagine you entering a classroom. In just one glance, you can tell:

- There are 30 students.
- One teacher is standing.
- There is a whiteboard.
- There are 10 benches.

->You don't look at each object separately. You understand the whole scene at once.

->YOLO works in a similar way.

Example

Imagine this image:



Car Person Bicycle

YOLO outputs:

[Car] Confidence: 98%

[Person] Confidence: 95%

[Bicycle] Confidence: 93%

->and draw a box around each object.

Where is YOLO Used?

Self-Driving Cars

Detect:

- Cars
- Pedestrians
- Traffic Lights
- Road Signs

CCTV Surveillance

Detect:

- People
- Suspicious activities
- Unauthorized entry

Medical Imaging

Detect:

- Tumors
- Abnormalities in X-rays

Retail Stores

Detect:

- Products on shelves
- Customer movement

YOLO vs LLM

LLM

Works with language.

Generates text.

Example: ChatGPT.

YOLO

Works with images and videos.

Detects and locates objects.

Example: Object detection in CCTV cameras.

11. Whisper

-> Whisper is an AI speech recognition model developed by OpenAI that converts spoken language (audio) into text.

Why is Whisper Needed?

-> Computers naturally understand text better than audio.

If you speak:

 "What is Artificial Intelligence?"

-> The AI first needs to convert your voice into text:

"What is Artificial Intelligence?"

Then the LLM (like ChatGPT) can understand and answer it.

How Does Whisper Work?

User Speaks



Audio Input



Whisper AI



Converts Speech to Text



LLM (ChatGPT)



Generates Response

Real-Life Applications



Voice Assistants: Siri, Google Assistant, ChatGPT Voice Mode



Video Subtitles: Automatically generate captions from spoken audio.



Lecture Notes: Convert a professor's lecture into text for students.



Call Centers: Convert customer calls into searchable text records.

12.Lang Chain

-> LangChain is a framework used by developers to build AI applications by connecting an LLM with tools like memory, RAG, databases, web search, and APIs. It acts like a bridge that connects all the AI components together.

Example

Suppose you want to build a College AI Assistant.

User asks:

"What is the attendance rule?"



LangChain receives the question



Retrieves information from the college PDF database (RAG)

↓
Sends the relevant information to the LLM
↓
LLM generates the answer
↓
Shows the response to the user.

->Here, LangChain connects the LLM with the college documents and tools to build the AI application.

13.CrewAI

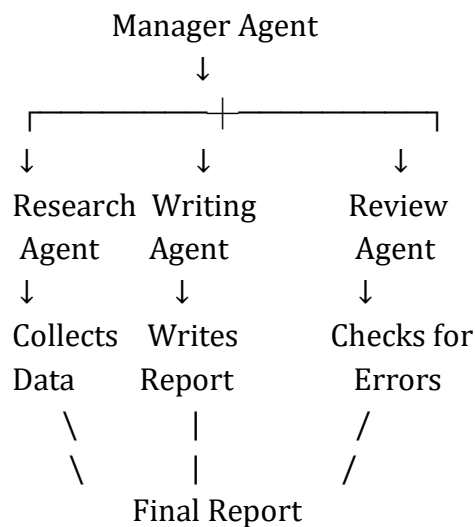
-> CrewAI divides a large task into smaller subtasks and assigns them to different AI agents that are better suited for those tasks. Each AI agent has its own role, and together they complete the overall objective.

Example

Suppose the goal is:

"Create a report on Artificial Intelligence."

->CrewAI may divide the work like this:



->Each AI agent performs the task it is best at, and together they complete the project.

AI Development Platforms

14.Github

-> GitHub itself does not create or train AI models. It mainly acts as an online storage and collaboration platform for code. It is like Google Drive, but instead of storing normal files, it is specially designed for storing and managing programming code and software projects.

Example:

- >Meta releases the open-source code for a project.
- >They upload or maintain it on GitHub.
- >A developer can visit GitHub, download (clone) the code, modify it, and build upon it.

Developer writes code



Uploads to GitHub



GitHub stores and manages the code



Other developers can view, download, or contribute

Real-Life Example

- >Suppose Meta releases a sample AI chatbot project.

Meta



Uploads AI chatbot code to GitHub



You download the code



You modify it for your college project



You upload your improved version to your own GitHub repository

15.Hugging Face

-> Hugging Face provides pre-trained AI models that developers can download, test, and fine-tune for their own applications instead of building and training a model from scratch. GitHub mainly stores the code of the application, while Hugging Face mainly stores the AI models.

Example

Suppose you want to build a medical chatbot:

Need a Medical AI



Download a pre-trained Llama model from Hugging Face



Fine-tune it with medical data



Write the application code in Python



Store and manage the code on GitHub

Easy Memory Trick

GitHub



Code Storage & Collaboration

Hugging Face



AI Model Storage, Testing & Fine-Tuning

16.Ollama

-> Ollama is a tool that allows you to download and run open-source AI models locally on your own computer.

Why is Ollama Needed?

-> Suppose you download a pre-trained Llama model from Hugging Face.

-> Now you need a way to actually **load and run that model** on your computer.

-> That's where Ollama comes in.

Download AI Model



Ollama



Runs the Model on Your Computer



You can chat with it or build apps using it

Example

-> Suppose you want to build your own local AI chatbot.

Step 1: Install Ollama.



Step 2: Download the Llama model.



Step 3: Ollama runs the model on your PC.



Step 4: Build your chatbot using Python or LangChain.

Vibe coding

17. AI Assisted Development

-> AI-Assisted Development is the process of using AI tools to help developers write, understand, debug, and improve code.

(or)

-> Instead of writing every line of code yourself, you work together with an AI assistant that helps you code faster and easier.

Why is it Needed?

-> Imagine you want to build a website login page.

Without AI:

1. Think of the code.
2. Search Google for syntax.
3. Write the code.
4. Debug errors.
5. Fix mistakes.

With AI-Assisted Development:

1. Tell the AI what you want.
2. AI generates a code draft.
3. You review and modify it.
4. AI helps fix bugs and explain errors.

->So, AI becomes like a **coding assistant**, not a replacement for the developer.

Example

Developer:

"Create a login page using HTML and CSS."

↓

AI Coding Assistant:

Generates the basic HTML and CSS code.

↓

Developer:

Reviews, edits, and adds custom features.

↓

Final Website is built faster.

18.AI Coding Tools

->AI Coding Tools are software applications that use AI to help developers write, understand, debug, and improve code.

Popular AI Coding Tools

1. GitHub Copilot

->AI coding assistant integrated into editors like VS Code.

->Suggests code as you type.

->Can generate functions and complete blocks of code.

2. ChatGPT

->Can generate code, explain concepts, debug errors, and help learn programming.

3. Claude

->AI assistant from Anthropic that can help with coding and large code analysis.

4. Gemini

->Google's AI assistant that also supports programming tasks.

5. Cursor

->An AI-powered code editor with built-in AI assistance.

Cursor

->Cursor is an AI-powered code editor that helps developers write, edit, understand, and debug code using AI directly inside the coding environment.

Why is Cursor Needed?

Normally, a developer might do this:

Write code in VS Code



Open ChatGPT in browser



Copy the code



Ask for help



Copy the answer back

With Cursor:

Write code in Cursor



Ask the built-in AI directly



AI explains, fixes, or generates code



Continue coding

->Everything happens in one place.

Example

->Suppose you are building a calculator website.

You type: **"Create a responsive calculator using HTML, CSS, and JavaScript."**

Cursor can:

1. Generate the initial code.
2. Create separate HTML, CSS, and JS files.
3. Help fix errors if the app doesn't work.
4. Explain any part of the code when you ask.

